

ACP-Gini: An Ancestor-Correlation Penalized Splitting Criterion for Multicollinearity-Robust Decision Trees

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Abstract

Decision trees avoid coefficient-identification problems associated with linear regression, but correlated predictors can still make their selected feature sets redundant and their importance profiles difficult to interpret. We introduce ACP-Gini, a single-tree splitting criterion that multiplies each candidate’s raw Gini gain by a product of correlation penalties against distinct features already used on the root-to-node path. The implementation reduces exactly to CART at $\alpha = 0$, excludes self-penalties, and uses raw gain for stopping. Evaluation combines a controlled correlated-feature generator with five real binary-classification data sets, 15 repeated cross-validation folds, and 50 bootstrap refits per fold. At selected α , ACP-Gini reduces bootstrap weighted redundancy by 12.1% on WDBC, 29.6% on Wine, 12.9% on Ionosphere, and 10.8% on Sonar, while predictive differences from CART are not statistically significant. A real-data operating sweep shows that WDBC at $\alpha = 0.6$ reduces weighted redundancy by 38.5% over three seeds with a 0.17 percentage-point accuracy change; the seed-42 five-fold result is 46.5% with identical accuracy. Synthetic group coverage and within-group concentration increase from $\alpha = 0$ to 1 at every tested correlation level, with a small rise in noise importance. Stability improves on Wine and Pima but decreases on WDBC and Sonar, revealing a stability–redundancy tradeoff rather than a uniform stability gain.

Keywords: Decision trees, multicollinearity, feature redundancy, model stability, interpretable machine learning

1 Introduction

Decision trees are often described as “immune” to multicollinearity because a split uses one predictor at a time. This claim is narrower than it first appears. Trees do avoid simultaneous coefficient estimation, yet greedy induction must still choose among correlated variables with nearly interchangeable gains. Small sample changes can alter the

representative chosen from a correlated block, while a single fitted tree may repeatedly use several members of the same block along different branches. Consequently, prediction can remain strong while the selected feature set is internally redundant and impurity-based importance is spread across substitutes.

The distinction matters for interpretable machine learning. A model can be accurate but give a needlessly repetitive account of the evidence used for its decisions. Correlated-feature effects on variable importance are well documented for forests [11, 12, 15], while tree instability motivated resampling methods such as bagging [1]. For an auditable single tree, however, preprocessing by variance-inflation-factor (VIF) pruning discards variables globally, and regularized-tree methods apply feature-level penalties that do not express the pairwise relationship between a candidate and the path that leads to the current node.

ACP-Gini addresses this local redundancy problem. At a node T , it discounts the Gini gain of candidate X_j according to its absolute correlation with distinct features used by ancestors of T . The design retains the familiar CART induction procedure [2, 8], remains exactly backward-compatible at $\alpha = 0$, and leaves raw gain responsible for deciding whether an informative split exists.

We evaluate four research questions. **RQ1:** Does ACP-Gini improve bootstrap stability of feature choice and importance relative to CART? **RQ2:** Does it reduce selected-set redundancy and concentrate importance on one representative per correlated group? **RQ3:** What accuracy tradeoff does α induce, and is the selected setting statistically distinguishable from CART? **RQ4:** How does it affect tree complexity, and how should interpretive compactness be characterized?

This study makes four contributions:

- an ancestor-correlation penalty for a single interpretable tree, including explicit no-self-penalty and raw-gain stopping rules;
- two direct measures of selected-set redundancy, plus bootstrap stability and split-composition measures;
- controlled group-level synthetic evaluation and a five-data-set real benchmark against CART, VIF+CART, and an RRF-inspired single tree;
- a reproducible from-scratch implementation whose tables and figures are generated from committed CSV outputs.

The scope is interpretable model construction and model diagnostics, complementing the broader argument for directly usable interpretable models in consequential settings [14].

2 Related Work

CART and instability.

CART greedily maximizes an impurity decrease at each node [2]. Greedy induction is sensitive to perturbations because an early alternative split changes all descendants. Li and Belford studied instability across decision-tree induction algorithms [10]; bagging was motivated partly by exploiting the variance of unstable learners [1]. These results concern predictive variance and structure, but do not directly regulate correlation among selected split features.

Table 1 Positioning of the evaluated single-tree approaches.

Method	Single tree	Corr.-aware	Path-local	No preproc.	CART at $\alpha = 0$
VIF+CART	Yes	Yes	No	No	No
RRF-style	Yes	No	No	Yes	No
ACP-Gini	Yes	Yes	Yes	Yes	Yes

Correlated predictors and importance.

Conditional permutation importance was proposed to address dependencies among predictors [15]. Nicodemus et al. demonstrated that correlation changes permutation-importance behavior [12]; Genuer et al. examined variable selection in random forests [7]; and Louppe et al. analyzed impurity-based importance in randomized trees [11]. These approaches are mainly diagnostic or ensemble-oriented. ACP-Gini changes the construction of one tree and retains unpenalized impurity gain for comparable feature-importance accounting.

Feature regularization and preprocessing.

Regularized Random Forest penalizes the gain of newly introduced features [4], and guided RRF incorporates external importance information [5]. Our RRF-style baseline is explicitly a single-tree adaptation so that all splitting methods share one implementation. VIF pruning represents the preprocessing family; the threshold of 10 follows common practice summarized by Dormann et al. [6]. Unlike global removal, ACP-Gini can retain a correlated predictor for a branch where it supplies useful conditional information.

To our knowledge, the gap is therefore specific: prior work does not inject an interpretable pairwise correlation with root-to-node ancestors into the split criterion of a single tree. ACP-Gini is not an ensemble replacement; it is a local criterion for reducing repetitive feature use while preserving the mechanics and inspectability of CART.

3 Methodology

Let T contain n_T samples and let T_L, T_R be a candidate split. With class proportions $p_k(T)$, Gini impurity and raw gain are

$$G(T) = 1 - \sum_k p_k(T)^2, \quad (1)$$

$$\Delta G(X_j, t) = G(T) - \frac{n_L}{n_T} G(T_L) - \frac{n_R}{n_T} G(T_R). \quad (2)$$

For each feature, the best threshold is found by sorting its values and evaluating midpoints between distinct adjacent values using cumulative class counts.

Let $A(T)$ be the set of distinct feature indices used by ancestors on the root-to- T path. ACP-Gini excludes the candidate itself from the penalty set:

$$A_j(T) = \{p \in A(T) : p \neq j\}, \quad (3)$$

$$P(X_j, T) = \prod_{p \in A_j(T)} (1 - \alpha |r(X_j, X_p)|), \quad (4)$$

$$S(X_j, t) = \Delta G(X_j, t)P(X_j, T). \quad (5)$$

The selected candidate maximizes $(S, \Delta G, -j)$ lexicographically. Thus ties first favor larger raw gain and then the lowest feature index. Repeated splits on the same continuous predictor are not spuriously multiplied by $(1 - \alpha)$.

Properties.

For fixed raw gain, the score is non-increasing in each absolute ancestor correlation and in α . The product accumulates moderate redundancy across several distinct ancestors. For $\alpha < 1$, all factors remain positive, so no feature is excluded by a hard threshold. At $\alpha = 0$, every factor equals one and the candidate ordering is exactly CART’s ordering. Importantly, stopping depends only on whether any candidate has positive raw gain; the penalty reorders informative candidates but never turns an informative node into a leaf.

Pearson correlation is computed once on the full training fold by default. Spearman correlation rank-transforms columns before the same calculation. Constant-column NaNs are replaced by zero correlation. A node-local ablation recomputes correlations when at least 30 samples reach the node and otherwise falls back to the global matrix. Global preprocessing costs $O(nd^2)$ once; lookup of ancestor correlations adds $O(d|A(T)|)$ per node beyond the threshold scan.

Parameter selection.

Inner stratified folds evaluate $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. Eligible values have macro-F1 within one standard deviation of the $\alpha = 0$ mean, and importance rank correlation selects among eligible values. The final data-set value is the lower quartile of fold-level selections, rounded down to the grid. We treat this data-set-level value as a conservative consensus operating point aggregated from per-fold inner-CV selections. It is not a fully nested unbiased selection protocol; however, the operating sweep in Fig. 4 shows that accuracy varies by at most approximately 0.3 percentage points on WDBC through $\alpha = .8$ and 0.2 points across the entire Wine grid. The documented WDBC $\alpha = 1$ failure is outside the selected operating region. This flat practical region makes material inflation of the reported selected-setting accuracy unlikely. The RRF-style $\lambda \in \{0.5, 0.7, 0.9\}$ uses the upper quartile because larger λ is the weaker penalty.

4 Experimental Setup

We use WDBC from scikit-learn [13], Wine Quality, Ionosphere, and Sonar from UCI [9], and Pima Diabetes from OpenML. All are numeric binary-classification tasks. Pairwise $|r| > 0.7$ occurs most frequently in WDBC and Sonar; Pima serves as a low-collinearity control. Cached CSV files accompany the code.

Experiments ran with Python 3.10.11 on 64-bit Microsoft Windows 11 Pro using an AMD Ryzen 7 PRO 5850U CPU. The complete experiment suite required approximately 35 wall-clock minutes on this machine.

Synthetic data contain five independent Gaussian signals with weights $(1, .8, .6, .4, .2)$, three correlated copies per signal, and five noise features. A copy is $\rho Z + \sqrt{1 - \rho^2}\epsilon$ for $\rho \in \{.7, .8, .9, .95, .99\}$. Group coverage is the fraction of the five

Table 2 Data-set statistics and collinearity diagnostics.

Data set	Samples	Features	Positive rate	Pairs with $ r > 0.7$
WDBC	569	30	0.627	0.161
Wine	6497	11	0.633	0.018
Ionosphere	351	34	0.359	0.005
Sonar	208	60	0.534	0.031
Pima	768	8	0.349	0.000

signal groups represented among the top-five importance features. Group concentration averages $\max_{j \in g} I_j / \sum_{j \in g} I_j$ over all five groups; a zero-mass group contributes zero. Noise importance is the total mass on the five noise columns.

All main methods use depth 6 and minimum leaf size 5. Three shuffled five-fold runs (seeds 42–44) provide 15 paired observations. Each training fold is additionally bootstrapped 50 times. Predictive metrics are accuracy, macro-F1, and AUC. Stability metrics comprise Jaccard similarity of all used-feature sets, top-five-importance Jaccard, Spearman correlation of importance vectors, and split-composition distance, defined as one minus multiset Jaccard over (depth, feature) internal-node pairs.

For a fitted tree with used set U , selected-set redundancy is the mean $|r_{ij}|$ over pairs in U . Importance-weighted redundancy is

$$R_w = \frac{\sum_{i < j} I_i I_j |r_{ij}|}{\sum_{i < j} I_i I_j}. \quad (6)$$

Both use the full-data correlation matrix for a consistent diagnostic scale and are reported for main fits and averaged bootstrap fits. The full-data correlation matrix serves exclusively as a label-free, post-hoc diagnostic scale and does not participate in model fitting or hyperparameter selection. Wilcoxon signed-rank tests use paired fold observations [3]. The four ACP-Gini-versus-CART bootstrap weighted-redundancy comparisons on the collinear data sets are the primary confirmatory family; all other Wilcoxon results are exploratory or descriptive. Repeated-CV folds share training data, so their p-values are approximate; we therefore emphasize effect sizes and cross-seed consistency. Root agreement is retained in raw CSVs but omitted from headline tables because ACP-Gini is identical to CART at the root by construction.

5 Results and Discussion

5.1 Correctness and Synthetic Ground Truth

The $\alpha = 0$ sanity suite passed all 15 seed–data-set cases. Pearson and Spearman modes produced identical structures and predictions when the penalty was inactive; held-out accuracy differed from scikit-learn’s tree by at most 0.01.

Figure 1 reports group-level results. At $\rho = .9$, group coverage increases from .68 to .74 and group concentration from .690 to .745 between $\alpha = 0$ and 1. Across the five ρ values, endpoint coverage gains range from .04 to .10 and endpoint concentration gains from .027 to .098. Intermediate values are not strictly monotone for every ρ , but every endpoint improves. Figure 2 shows that accuracy remains near .79 at $\rho = .9$, while noise importance rises from .018 to approximately .033. This is a small but real side effect: discouraging redundant signal variables occasionally exposes weak noise splits.

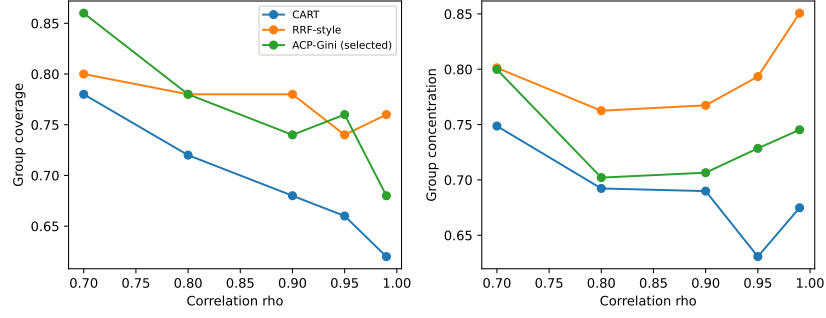


Fig. 1 Synthetic group coverage and concentration. RRF-style attains high concentration by rote feature reuse; see Sect. 5.2 for why this does not transfer to real data.

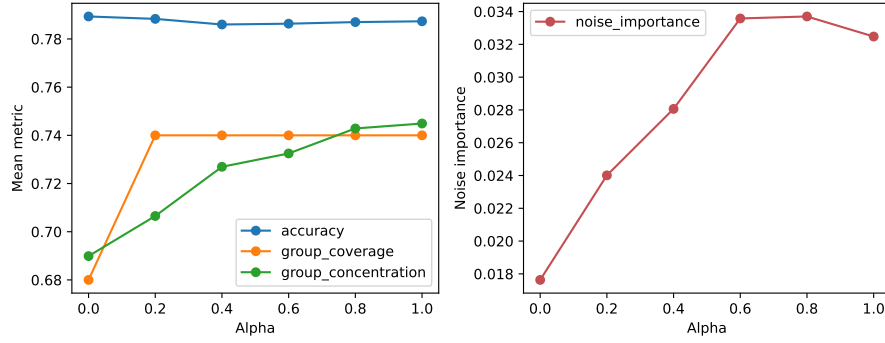


Fig. 2 ACP-Gini sensitivity at $\rho = .9$: accuracy and group metrics (left), and the noise-importance side effect (right).

RRF-style also attains strong synthetic concentration because its fixed penalty rewards rote reuse of already-used features, without considering correlation strength. This behavior does not transfer to real data: bootstrap weighted redundancy is .603 versus CART’s .621 on WDBC and .181 versus .183 on Wine, while ACP-Gini reaches .545 and .128. The contrast motivates a correlation-aware penalty rather than a fixed new-feature penalty.

Exact-original recovery is less suitable at high ρ : an original and its copy are close substitutes, and selecting one representative should not be counted as failure merely because it is not the designated original column. The group metrics align the evaluation with the method’s intended behavior.

5.2 Prediction and Selected-Set Redundancy

The conservative selected values are .2 (WDBC), 1.0 (Wine), .4 (Ionosphere), .2 (Sonar), and .2 (Pima). Accuracy differences from CART are statistically indistinguishable on every data set. Sonar’s direction is negative but shrinks to only .65 percentage points after lower-quartile aggregation; Ionosphere changes from .885 to .886.

The primary result is lower internal correlation. Bootstrap weighted redundancy falls from .621 to .545 on WDBC (12.1%), .183 to .128 on Wine (29.6%), .168 to .146 on Ionosphere (12.9%), and .218 to .195 on Sonar (10.8%). The four primary comparisons all have $p \leq .0003$ and remain significant under Holm correction at $\alpha = .05$. Pima changes by only .0017 in absolute terms; despite a small exploratory p-value

Table 3 Selected data-set-level penalty parameters.

Data set	α	λ
Ionosphere	0.400	0.900
Pima	0.200	0.900
Sonar	0.200	0.900
WDBC	0.200	0.700
Wine	1.000	0.900

Table 4 Predictive results over 15 held-out folds.

Data set	Method	Accuracy (mean)	Accuracy (std.)	Macro-F1 (mean)	Macro-F1 (std.)	AUC (mean)	AUC (std.)
Ionosphere	ACP-Gini	0.886	0.028	0.876	0.030	0.904	0.036
Ionosphere	CART	0.885	0.018	0.874	0.020	0.910	0.035
Ionosphere	RRF-style	0.885	0.021	0.875	0.022	0.910	0.035
Ionosphere	VIF+CART	0.885	0.018	0.874	0.020	0.910	0.035
Pima	ACP-Gini	0.733	0.028	0.692	0.036	0.774	0.034
Pima	CART	0.737	0.030	0.698	0.036	0.774	0.040
Pima	RRF-style	0.735	0.030	0.696	0.035	0.776	0.035
Pima	VIF+CART	0.737	0.030	0.698	0.036	0.774	0.040
Sonar	ACP-Gini	0.731	0.075	0.729	0.075	0.764	0.078
Sonar	CART	0.737	0.065	0.736	0.066	0.752	0.080
Sonar	RRF-style	0.726	0.078	0.724	0.078	0.750	0.086
Sonar	VIF+CART	0.715	0.048	0.712	0.049	0.754	0.060
WDBC	ACP-Gini	0.934	0.024	0.929	0.027	0.957	0.020
WDBC	CART	0.934	0.018	0.929	0.020	0.953	0.020
WDBC	RRF-style	0.941	0.028	0.936	0.032	0.958	0.018
WDBC	VIF+CART	0.931	0.025	0.926	0.028	0.962	0.021
Wine	ACP-Gini	0.746	0.011	0.726	0.011	0.800	0.008
Wine	CART	0.747	0.011	0.725	0.012	0.797	0.012
Wine	RRF-style	0.748	0.010	0.724	0.012	0.799	0.012
Wine	VIF+CART	0.747	0.011	0.724	0.012	0.798	0.011

Table 5 Selected-set redundancy; lower is better.

Data set	Method	Set red. (main)	Weighted red. (main)	Set red. (boot.)	Weighted red. (boot.)
Ionosphere	ACP-Gini	0.155	0.157	0.159	0.146
Ionosphere	CART	0.169	0.169	0.183	0.168
Ionosphere	RRF-style	0.175	0.168	0.183	0.166
Ionosphere	VIF+CART	0.169	0.169	0.182	0.167
Pima	ACP-Gini	0.168	0.182	0.169	0.178
Pima	CART	0.169	0.183	0.170	0.180
Pima	RRF-style	0.167	0.182	0.169	0.180
Pima	VIF+CART	0.169	0.183	0.170	0.180
Sonar	ACP-Gini	0.208	0.193	0.204	0.195
Sonar	CART	0.238	0.228	0.234	0.218
Sonar	RRF-style	0.228	0.225	0.226	0.216
Sonar	VIF+CART	0.213	0.185	0.220	0.195
WDBC	ACP-Gini	0.345	0.572	0.338	0.545
WDBC	CART	0.433	0.634	0.465	0.621
WDBC	RRF-style	0.448	0.613	0.458	0.603
WDBC	VIF+CART	0.330	0.460	0.283	0.343
Wine	ACP-Gini	0.217	0.118	0.220	0.128
Wine	CART	0.244	0.166	0.245	0.183
Wine	RRF-style	0.243	0.165	0.244	0.181
Wine	VIF+CART	0.238	0.158	0.238	0.169

from repeated observations, this is no meaningful reduction and supports its role as a low-collinearity control. Figure 3 visualizes the comparison.

VIF+CART reaches lower WDBC redundancy (.343 versus ACP-Gini’s .545) by globally deleting 13 of 30 features before training. That reduction entails irreversible information loss and a fixed feature set; ACP-Gini keeps every feature available for branch-conditional use at equal accuracy.

The operating sweep in Fig. 4 separates a conservative default from attainable operating points. Across all three seeds, WDBC at $\alpha = .6$ reduces main-fit weighted redundancy from .634 to .390 (38.5%) while accuracy changes from .934 to .932.

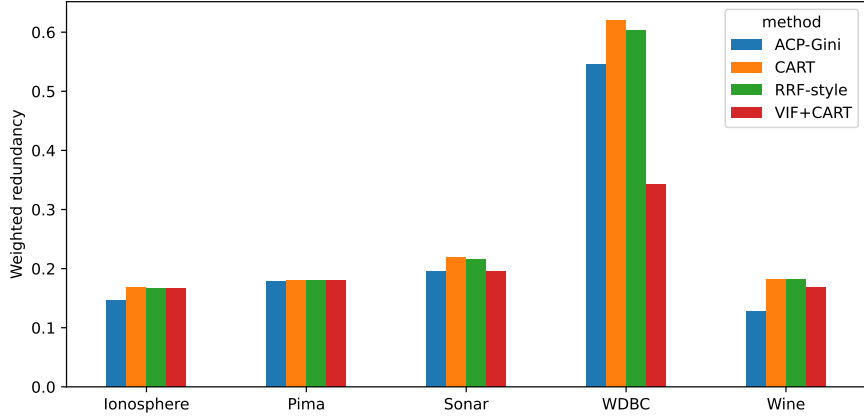


Fig. 3 Mean bootstrap importance-weighted redundancy. Lower values indicate less internally correlated selected evidence.

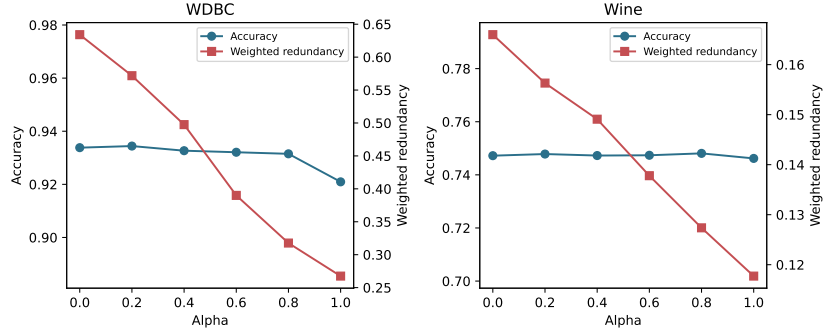


Fig. 4 Real-data operating characteristics for WDBC and Wine. Accuracy uses the left axis and weighted redundancy the right axis.

The pre-specified seed-42 five-fold check reproduces .656 to .351 (46.5%) with accuracy unchanged at .926. Wine shows a smooth redundancy decline from .166 to .118 while accuracy stays between .746 and .748. Thus practitioners can increase α when redundancy reduction is worth a small tolerance in predictive performance.

5.3 The Stability–Redundancy Tradeoff

Importance rank correlation rises by .179 on Wine and .017 on Pima; top-five Jaccard rises by .069 and .010, respectively. Wine’s gains are significant at $p = .0001$. Conversely, rank stability decreases by .057 on WDBC and .040 on Sonar, with smaller negative changes on Ionosphere. Table 6 shows this conditional pattern.

This is not equivalent to predictive failure. CART’s apparent stability can arise from repeatedly selecting the same correlated block, whereas ACP-Gini deliberately diversifies downstream evidence. Lower redundancy and higher cross-bootstrap rank agreement are distinct goals. ACP-Gini improves both where redundant choice drives instability (Wine), but trades rank consistency for less correlated feature use in block-correlated WDBC and Sonar.

5.4 Ablation, Runtime, and Qualitative Analysis

The ablation does not identify a universally dominant configuration. Spearman obtains the highest WDBC accuracy (.937), while Pearson-min is strongest on synthetic

Table 6 Bootstrap stability metrics.

Data set	Method	Feature-set	Jaccard	Top-5 Jaccard	Importance rank corr.	Split-comp. distance
Ionosphere	ACP-Gini		0.322	0.293	0.372	0.880
Ionosphere	CART		0.335	0.304	0.388	0.878
Ionosphere	RRF-style		0.319	0.304	0.378	0.882
Ionosphere	VIF+CART		0.338	0.305	0.393	0.877
Pima	ACP-Gini		0.899	0.677	0.714	0.663
Pima	CART		0.928	0.668	0.698	0.664
Pima	RRF-style		0.862	0.671	0.701	0.669
Pima	VIF+CART		0.928	0.668	0.698	0.664
Sonar	ACP-Gini		0.122	0.106	0.088	0.953
Sonar	CART		0.149	0.111	0.128	0.947
Sonar	RRF-style		0.128	0.111	0.112	0.952
Sonar	VIF+CART		0.166	0.128	0.155	0.945
WDBC	ACP-Gini		0.268	0.224	0.241	0.908
WDBC	CART		0.306	0.250	0.298	0.899
WDBC	RRF-style		0.246	0.237	0.265	0.901
WDBC	VIF+CART		0.311	0.267	0.308	0.884
Wine	ACP-Gini		0.898	0.675	0.845	0.512
Wine	CART		0.986	0.606	0.666	0.568
Wine	RRF-style		0.953	0.603	0.666	0.573
Wine	VIF+CART		0.993	0.620	0.766	0.549

Table 7 Correlation, aggregation, and scope ablation.

Data set	Correlation	Aggregation	Scope	Accuracy	Macro-F1	Fit time (s)
Synthetic	pearson	mean	global	0.785	0.784	0.087
Synthetic	pearson	min	global	0.796	0.796	0.085
Synthetic	pearson	product	global	0.792	0.792	0.092
Synthetic	pearson	product	node	0.793	0.792	0.087
Synthetic	spearman	product	global	0.792	0.791	0.092
WDBC	pearson	mean	global	0.935	0.930	0.033
WDBC	pearson	min	global	0.933	0.929	0.031
WDBC	pearson	product	global	0.933	0.928	0.031
WDBC	pearson	product	node	0.932	0.927	0.031
WDBC	spearman	product	global	0.937	0.934	0.033

Table 8 Measured fit time by data set and method.

Data set	Method	mean	std
Ionosphere	ACP-global	0.041	0.007
Ionosphere	ACP-node	0.044	0.013
Ionosphere	CART	0.042	0.011
Pima	ACP-global	0.025	0.000
Pima	ACP-node	0.029	0.000
Pima	CART	0.022	0.001
Sonar	ACP-global	0.079	0.019
Sonar	ACP-node	0.068	0.005
Sonar	CART	0.068	0.011
WDBC	ACP-global	0.038	0.005
WDBC	ACP-node	0.037	0.002
WDBC	CART	0.044	0.005
Wine	ACP-global	0.092	0.017
Wine	ACP-node	0.100	0.022
Wine	CART	0.069	0.005

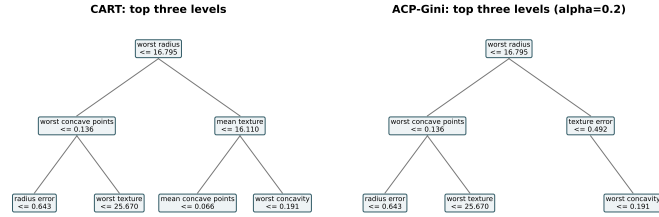
accuracy (.796); product-Pearson-global remains the simplest pre-specified default. Global-scope runtime ranges from .85 to 1.33 times CART across data sets, and node-local scope from .84 to 1.44 times. The overhead is data dependent and is not uniformly below 10%.

Figure 5 illustrates that the WDBC root and left branch remain identical, while the right child changes from mean texture to texture error under ACP-Gini. Average node

Table 9 Tree complexity at the selected operating points.

Data set	Method	Nodes	Depth	VIF	dropped
Ionosphere	ACP-Gini	24.467	6.000		0.000
Ionosphere	CART	25.133	6.000		0.000
Ionosphere	RRF-style	24.867	6.000		0.000
Ionosphere	VIF+CART	25.133	6.000		1.000
Pima	ACP-Gini	71.000	6.000		0.000
Pima	CART	71.267	6.000		0.000
Pima	RRF-style	71.267	6.000		0.000
Pima	VIF+CART	71.267	6.000		0.000
Sonar	ACP-Gini	25.533	5.400		0.000
Sonar	CART	25.800	5.267		0.000
Sonar	RRF-style	25.933	5.200		0.000
Sonar	VIF+CART	26.467	5.133		17.067
WDBC	ACP-Gini	24.200	5.733		0.000
WDBC	CART	24.733	5.800		0.000
WDBC	RRF-style	25.133	5.733		0.000
WDBC	VIF+CART	27.933	5.533		13.267
Wine	ACP-Gini	115.800	6.000		0.000
Wine	CART	114.733	6.000		0.000
Wine	RRF-style	114.867	6.000		0.000
Wine	VIF+CART	115.000	6.000		1.000

counts remain comparable because raw-gain stopping preserves informative splitting: for example, WDBC uses 24.2 nodes for ACP-Gini and 24.7 for CART.

**Fig. 5** Top three levels of full-data WDBC trees at the selected $\alpha = .2$.

Across the WDBC repeated folds, CART and ACP-Gini disagree on 65 predictions. CART alone is correct in 33 cases and ACP-Gini alone in 32; 20 disagreements are near a $[.4, .6]$ probability boundary. The near balance is consistent with unchanged aggregate accuracy.

Research questions.

RQ1 is conditional: stability improves on Wine and Pima but decreases on WDBC and Sonar. RQ2 is supported by synthetic group metrics and the lower selected-set redundancy on four collinear real data sets. RQ3 is supported: selected settings have statistically indistinguishable accuracy, while the sweep exposes the adjustable trade-off. RQ4 is not supported as originally framed: node counts and depth are comparable; compactness is better characterized as lower redundancy of the selected evidence.

Limitations.

ACP-Gini is a greedy heuristic without a global optimality guarantee. Correlation does not distinguish redundancy from useful conditional interaction. High α can increase noise importance and, on WDBC at $\alpha = 1$, reduce accuracy to .921. Results concern numeric binary data and one tree; categorical association measures such as Cramér’s V and ensemble integration remain future work. The five data sets provide varied correlation structures but do not establish universal behavior.

6 Conclusion

ACP-Gini introduces an interpretable path-dependent correlation penalty while preserving CART behavior at $\alpha = 0$, repeated-feature splitting, and raw-gain stopping. The strongest evidence concerns redundancy rather than a uniform stability gain: selected feature sets are less internally correlated on four collinear real data sets without statistically significant predictive loss, and synthetic group-level recovery improves across all tested correlation endpoints. Stability depends on data structure, yielding an informative stability–redundancy tradeoff. Future work should study theoretical conditions for beneficial penalties, categorical association, regression trees, and ensemble variants.

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